ST.PHILOMENA'S COLLEGE (AUTONOMOUS), MYSURU

(AFFILIATED TO UNIVERSITY OF MYSORE)

REACCREDITED BY NAAC WITH A GRADE

Three-year six semesters Choice Based Credit System (CBCS) with
Learning Outcome Based Curriculum framework (LOCF)
And Continuous Assessment & Grading Pattern (CAGP) Undergraduate Programme Under Autonomous Structure

Programme- B.Sc.

The academic year 2018-19 onwards

DEPARTMENT OF MICROBIOLOGY

VISION AND MISSION OF THE COLLEGE

VISION:
The college is guided by the visionary zeal of providing value- based education to everyone irrespective of religion, caste, creed or sex by which the character is formed, intellect is explained and one can stand on his/her feet**.**

MISSION:
To transform young men and women who come to learn not from books, but also from life and to share the experience of working and playing together, this inculcates life skills to become good citizens with integrity and discipline.

Programme Educational Objective (PEO)

PEO-1	Graduates will be able to master and display competency and leadership to become successful professionals, employees and entrepreneurs or pursue higher education and research.
PEO-2	Graduates will be able to demonstrate the commitment towards professional ethics, gender sensitivity, preservation of environment and sustainable development.
PEO-3	Graduates will continue to learn and advance their careers through activities such as participation in professional organizations, attainment of professional certification and seeking higher education.

Programme Outcomes (PO): BSc. Programme

PO-1	**Disciplinary Knowledge:** The BSc. graduates will acquire the knowledge with facts and figures related to pure and applied sciences. Understand the basic concepts, fundamental principles, and the scientific theories related to various scientific phenomena and their relevancies in the day-to-day life.
PO-2	**Cognitive and Communicative skills:** Students learn two languages along with three major subjects. At the end of the programme, the students would have developed reading, writing, speaking, interpretive and composition skills. They would be able to communicate with others using appropriate media; confidently share one's views and express themselves
PO-3	**Research Related Skills:** The BSc. students will acquire the skills in handling scientific instruments, planning and performing in laboratory experiments.
PO-4	**Ethics**: The BSc. students will have imbibed ethical, moral and social values in personal and social life leading to highly cultured and civilized personality.
PO-5	**Problem Solving:**The BSc. graduates will develop the ability to analyze and solve Course-related problems and also the ability to evaluate situations and react responsibly to communicate, cooperate and lead a team among peers and others.

PO-6	**Critical Thinking:** The qualities of a science student – observation, precision, analytical mind, logical thinking, clarity of thought and expression, systematic approach, qualitative and quantitative decision making are enhanced.
PO-7	**Social Interaction:** The BSc. graduates shall appreciate the role of science in society; and its personal, social and global importance.
PO-8	**Analytical Skills**: The graduates will master the skills of observations and drawing logical inferences from the scientific experiments. They will be able to analyze the given scientific data critically and systematically and acquire the ability to draw the objective conclusions.
PO-9	**Environment and Sustainability:** Graduates will be able to understand the issues of environment and work towards sustainable development.
PO-10	**Employability:** After completing the programme, graduates will have the competency to be employed or to be an entrepreneur.
PO-11	**Leadership Quality:** In the graduation programme students are inculcated moral and ethical values, managerial skills, adoptability, problem solving, taking initiative, decision making, risk taking to make them confident leaders.

Programme Specific Outcomes (PSO)- BSc. Programme

PSO-No	After the completion of BSc. programme by studying Biotechnology, Biochemistry and Microbiology (BtBMb) students will be able to	Cognitive level
PSO-1	Demonstrate and apply their knowledge of cell biology, biochemistry, microbiology and molecular biology to solve the problems related to the field of B**iotechnology.**	Apply
PSO-2	To acquire in-depth theoretical and practical knowledge of Biochemistry and the ability to apply the acquired knowledge to provide cost efficient solutions in Biochemistry. And to bridge the gap between academia and industry	Understand &Apply
PSO-3	Apply the scientific method and hypothesis testing in the design and execution of experiments related to isolation, identification, cultivation and control of microorganisms from food, human body, environmental sources.	Understand and apply

Mapping of Mission of the College with PEO			
Mission	PEO-1	PEO-2	PEO-3
Mission -1	✓	✓	✓

Mapping of PEOs with Programme Outcocomes(PO)											
PEO No.	**PO-1**	**PO-2**	**PO-3**	**PO-4**	**PO-5**	**PO-6**	**PO-7**	**PO-8**	**PO-9**	**PO-10**	**PO-11**
PEO-1		✓	✓		✓					✓	✓
PEO-2	✓			✓					✓		
PEO-3						✓	✓	✓			

PREAMBLE

When the college became autonomous in the year 2011, the B.Sc. Microbiology syllabus of the University of Mysore was adopted with a few modifications for the academic year 2011-12. The second revision of the syllabus was done in the year 2013 wherein new topics were included resulting in about 30% revision. The third revision was done for the academic year 2016 onwards.

The existing syllabus has been revised for its implementation under the CBCS from the academic year 2018-19. The syllabus for hard core papers has been retained with minor changes. Syllabus for ten soft core papers was prepared after referring to syllabi of other universities, both national and international. The topics of the papers have been selected focusing on the latest developments in the core areas of Microbiology and with the purpose of skill development and enhancement. It was reviewed and refined in the BOS meeting with valuable inputs from the members.

CBCS-LOCF SYLLABUS FOR B.Sc.
THE ACADEMIC YEAR 2018-19 ONWARDS
GENERAL SCHEME WITH RESPECT TO TEACHING & EVALUATION

Discipline Specific Core (DSC) or Hard Core (HC) Papers

Semester	Title of the Paper	TYPE	Course	Teaching Hours per Week	Credits	Exam Duration	Max. Marks Theory/Practical		
				Theory/ Practical	Theory/ Practical	Theory/ Practical	Theory/Practical	I A Theory/Practical	Total Marks
I	Paper-I Title: General Microbiology	DSC	MA580	03	03	03	50	20	100
	Practical Paper-I	DSC	MA582	03	1.5	03	20	10	
II	Paper-II. Title: Microbial Diversity	DSC	MB580	03	03	03	50	20	100
	Practical Paper-II	DSC	MB582	03	1.5	03	20	10	
III	Paper-III. Title: Microbial Growth and Genetics	DSC	MC580	03	03	03	50	20	100
	Practical Paper-III	DSC	MC582	03	1.5	03	20	10	
IV	Paper-IV. Title: Microbial metabolism and Genetic Engineering	DSC	MD580	03	03	03	50	20	100
	Practical Paper-IV	DSC	MD582	03	1.5	03	20	10	
V	Paper-V. Title: Environmental Microbiology	DSC	ME580	03	03	03	70	30	300
	Paper-VI.Title: Soil and Agricultural Microbiology	DSC	ME582	03	03	03	70	30	
	Practical Paper-V.	DSC	ME584	03	1.5	03	35	15	
	Practical Paper-VI.	DSC	ME586	03	1.5	03	35	15	
VI	Paper-VII.Title: Food Microbiology and Industrial Microbiology	DSC	MF580	03	03	03	70	30	300
	Paper-VIII. Title: Immunology And Medical Microbiology	DSC	MF582	03	03	03	70	30	
	Practical Paper-VII.	DSC	MF584	03	1.5	03	35	15	
	Practical Paper-VIII.	DSC	MF586	03	1.5	03	35	15	
		DSE		02	02	03	30	20	100

		DSE 2		**02**	**02**	**03**	**30**	**20**	
					40	**-**	**760**	**340**	**1100**

Discipline Specific Elective (DSE or Soft Core (SC)

SL.No	**Title of the Paper**	**TYPE**	**Course Code**		**Semester**		**Examination Scheme**				
						Theory	**Credits**	**Exam Duration in Hours**	**Theory Max. Marks**	**I A Max Marks**	**Total Marks**
1.	**Analytical methods in Microbiology**	**DSE**	M58Y04		**II**	**2**	**2**	**03**	**30**	**20**	**50**
2.	**Applied Mycology**	**DSE**	M58Y02			**2**	**2**	**03**	**30**	**20**	**50**
3.	**Basics of Research Methodology**	**DSE**	M58Y09		**to**	**2**	**2**	**03**	**30**	**20**	**50**
4.	**Diagnostic Virology**	**DSE**	M58Y01		**IV**	**2**	**2**	**03**	**30**	**20**	**50**
5.	**Clinical Bacteriology**	**DSE**	M58Y05			**2**	**2**	**03**	**30**	**20**	**50**
6.	**Clinical Parasitology**	**DSE**	M58Y06		**V**	**2**	**2**	**03**	**30**	**20**	**50**
7.	**Plant Clinic**	**DSE**	M58Y10		**to**	**2**	**2**	**03**	**30**	**20**	**50**
8.	**Project Work With Seminar**	**DSE**	M58Y08		**VI**	**2**	**2**	**03**	**30**	**20**	**50**
9.	**Environmental Microbiology and Public Health**	**DSE**	M58Y03			**2**	**2**	**03**	**30**	**20**	**50**
10.	**Marine Microbiology**	**DSE**	M58Y07			**2**	**2**	**03**	**30**	**20**	**50**

Note:

Sl. No.	**Type**	
1.	**DSC or HC**	**Discipline Specific Core (DSC) or Hard Core (HC)**
2.	**DSE or SC**	**Discipline Specific Elective (DSE or /Soft Core (SC)**
3.	**SEC or OE**	**Skill Enhancement Course (SEC) or Open Elective (OE)**

SYLLABUS FOR B.SC. PROGRAMME UNDER CBCS LOCF SCHEME
DURATION OF THE COURSE - THREE YEARS-SIX SEMESTERS
FROM THE ACADEMIC YEAR- 2018-2019 Onwards

FIRST SEMESTER
MICROBIOLOGY PAPER-I
Title: GENERAL MICROBIOLOGY
Class duration – 03 hours per week=48 Hrs.
Marks: Theory (50) + Internal Assessment (20) = 70

Course Objectives (CO):

1. To get introduced to the field of Microbiology and its historical development, basic and applied aspects of Microbiology, scope of Microbiology.

2. To learn the basic microbiological techniques viz. microscopy, staining and sterilization involved in the handling, control and study of microorganisms.

3.To learn and understand the working principles of the light microscopy and to understand their applications of the principles in the laboratory.

Course Learning outcome(CO):

Course Learning Outcome:

CO	After the completion of this course, the student will be able to	Cognitive level
CO-01	To describe the milestones in the historical development of Microbiology	Understand
CO-02	To understand the use of physical and chemical methods of sterilization	Understand
CO-03	To demonstrate theoretical and practical skills in microscopy and their handling techniques and staining procedures	Apply

Unit I	**History of Microbiology:**	
1.1	Milestones in the historical development of Microbiology. Discovery of microorganisms, contributions of Antony van Leeuwenhoek. Theory of spontaneous generation and biogenesis. Contributions of Edward Jenner, Louis Pasteur, Joseph Lister, Robert Koch, Metchnikoff, Beijerinck, Winogradsky, Ivanowsky, Alexander Fleming, Selman Waksman (in brief) to the development of microbiology. Recent developments in the field of Microbiology.	**10 hrs**
1.2	Branches of Microbiology.	**1hrs**
1.3	Scope of Microbiology.	**1hrs**
Unit II	**Light Microscopy:**	
2.1.	Different types of microscopes, their construction and working principles. Simple microscope (dissection microscope).	**2hrs**

2.2.	Compound microscope (types of microscopy: bright field, dark field, phase contrast and fluorescence). Stereo microscope.	**6hrs**
Unit III	**Electron Microscopy:**	
3.1	The principle, construction and mode of operation of scanning and transmission electron microscope, limitations.	**2hrs**
3.2	Preparation of specimen for electron microscopic studies (ultra-thin sectioning, negative staining, shadow casting and freeze etching – in brief). Confocal microscope, Scanning probe microscope.	**3hrs**
Unit IV	**Microbiological stains and staining techniques:**	
4.1	Types of stains and principles of staining. Preparations of bacterial smears for light microscopy: Fixation.	**2hrs**
4.2	Simple staining (direct and indirect), differential staining (Gram's staining and acid-fast staining), Structural staining (capsule, flagella, cell wall and endospore of bacteria, nuclear staining). Wet mounting of algae and fungi. Hanging drop technique.	**5hrs**
Unit V	**Principles and methods of Sterilization :**	
5.1 5.1.1	**Physical methods and their mode of action** Heat- Dry heat – Hot air Oven, Incineration. Moist heat – Autoclave, Pressure cooker, Tyndallization (fractional sterilization). Definition of terms- TDT, TDP, D value, z value.	**5hrs.**
5.1.2	Filtration- Types of filters (membrane filters, Sietz filter, sintered glass filter, diatomaceous earth filter and Chamberland filter, HEPA-Laminar airflow system).	**4hrs.**
5.1.3	Radiation methods – UV radiation, γ rays and cathode rays.	**2hrs.**
5.2	**Chemical methods:** Definition of terms – disinfectants, antiseptics, sanitizers, microbicides – bactericide, virucide, fungicide, and sporicide, micro biostatic – bacteriostatic and fungistatic agents.	**2hrs.**
5.3	Use and mode of action of – alcohols, aldehydes, halogens, phenols, heavy metals, detergents: quaternary ammonium compounds.	**3hrs.**
REFERENCES:		

1. Ananthanarayanan, C. and Paniker, C.K.J. 2006. **Text Book of Microbiology**, Seventh Edn. Orient Longman Ltd., Chennai.
2. Aneja, K.R. 1993. **Experiments in Microbiology, Plant Pathology**. Rastogi and Company, Meerut.
3. Brock T.D. 2012. **Biology of Microorganisms,** Thirteenth Edn. Prentice-Hall Publications.
4. Cappuccino, J.G. and Sherman, N. 2004. **Microbiology-A Laboratory Manual**, Seventh Edition. Addison –Wesley.
5. Edward Alcamo. 2010. F**undamentals of Microbiology**, Ninth Edn. Jones and Barlett.
6. Salle, A.J. 1967. **Fundamental Principles of Bacteriology**, Sixth edition. Tata McGraw Hill Publishing Co. Ltd., New Delhi.
7. Kathleen Park Talaro 2009. **Foundations in Microbiology**, 7th International Edn, McGraw Hill.
8. Pelczar, Jr., J.M., Chan, E.C.S. and Kreig, N.R. 1993. **Microbiology**, Fifth Edn. Tata McGraw Hill Publishing Co. Ltd.
9. Prescott, L.M., Harley, J.P. and Klein, D.A. 2008. **Microbiology**, International Edn., Seventh Edn. WBC McGraw Hill.
10. Tortora, Funke, Chase. 2008. **Microbiology-An introduction**, 10th Edn. Pearson Education, Inc.
11. Veerakumari, L. 2006. **Bioinstrumentation**. MJP Publishers.

SEMESTER I

PRACTICAL I :GENERAL MICROBIOLOGY

Experiments based on theory syllabus 14 practical (One practical of 3 hrs/week)

Expt. No:

1. Laboratory safety: General rules and regulations.
   Study of simple and compound microscopes, their handling including oil immersion objective.
2. Preparation of stains ,mordant and mounting media – Methylene blue, crystal violet ,safranin ,nigrosin, carbol fuchsin, malachite green, Gram's iodine, cotton blue, glycerine, lactophenol.

4-7. Preparation of microorganisms for light microscopic observation-simple (direct and indirect) staining, differential staining (Gram's staining), structural staining (capsule, flagella and endospore of bacteria).

8. Staining and mounting of algae(e.g., *Spirogyra/Nostoc*) and fungi (e.g., b *Rhizopus/Aspergillus)*
9. Observation of bacterial motility by Hanging drop method.

10-11. Microscopic measurement of microorganisms/spores using stage and ocular micrometer.

12. Demonstration of laboratory equipment -autoclave, pressure cooker, hot air oven, incubator, inoculation hood/chamber, inoculation loop, inoculation needle, membrane filter and colony counter.
13. Preparation of chromic acid and its use. Use and mode of action of detergents, disinfectants- phenol, ethyl alcohol, formaldehyde, mercuric chloride and iodine.
14. Display of photographs of microbiologists and microscopes mentioned in the theory syllabus.

INTERNAL ASSESSMENT	**10 marks**
Break up	
Record	05 marks
Continuous assessment	05 marks

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SEMESTER- II
MICROBIOLOGY
Title: MICROBIAL DIVERSITY (PAPER-II)
Class Duration: 3 hours per week = 48 Hrs.
Marks: Theory (50) + Internal Assessment (20) = 70

Course Objectives (CO):

1.To get acquainted with the microbial world.

2. To study the characteristics of prokaryotes and eukaryotes and understand the structural similarities and differences among various groups of microorganisms

3. To study the cytology of microorganisms

4. To study the diversity and biology of microorganisms with reference to their structure, reproduction, classification, identification and economic importance.

Course Learning outcome (CO):

CO	**After the completion of this course, the student will be able to**	**Cognitive Level**
CO-01	To describe the characteristics of the groups of microorganisms.	Understand
CO-02	Identify and classify microorganisms based on their characteristics	Analyse
CO-03	To understand the methods of reproduction and economic importance of microorganisms	Understand

Unit I	**The Microbial World**:	
1.1	Groups of microorganisms: viruses, prokaryotes (cyanobacteria, bacteria, archaea), eukaryotes (algae, fungi, protozoa).	**2hrs**
1.2	Study of **a)**. Ultra structure of typical prokaryotic cell; the structure of bacterial cell –cell wall (Gram-positive, gram-negative, archaeobacteria), capsule, cell membrane (bacterial and archaeal), cytoplasm, ribosomes, nucleoid, plasmids, flagella, pili(fimbriae), inclusion bodies **b).** Ultrastructure of a eukaryotic cell. **c).** A comparative account of a prokaryotic and eukaryotic cell.	**5hrs**
1.3	General principles of classification and nomenclature of microorganisms (Haeckel's three kingdom classification and Whittaker's five kingdom classification), Carl Woese's domain system of classification.	**3hrs**
Unit II	**Viruses:**	
2.1	Definition, history of virology, general characteristics of viruses-size, shape and chemical composition, properties used for classification of viruses, isolation and	**3hrs**

	identification of viruses.	
2.2	Study of structure and replication of viruses: Phytophagenae-TMV. Zoophagenae-influenza and HIV Bacteriophages-T4 phage, λ phage Cyanophages	**5hrs**
2.3	Viroids, Prions and Virusoids. Importance of viruses.	**1 hr.**
Unit III	**Bacteria:**	
3.1.	Occurrence, shape and arrangement of bacterial cells, reproduction in bacteria, endospore formation. Classification of bacteria in brief as per Bergey's Manual of Systematic Bacteriology.	**2hrs**
3.2.	Study of the following bacteria: *E. coli, Staphylococcus aureus, Bacillus* spp. *Pseudomonas* spp.	**2hrs**
3.3	Study of the following in brief with examples: Rickettsiae, Chlamydias, Mycoplasmas, Spirochaetes, Actinomycetes, Archaea.	**3hrs**
3.4.	Study of Cyanobacteria: Occurrence, structure, reproduction of the following: *Microcystis, Spirulina*, and *Anabaena*.	**2hrs**
Unit IV	**Algae:**	
4.1	Distribution, the structure of the typical algal cell (E.g. *Chlamydomonas*).A brief account of reproduction.	**2hrs**
4.2	Study of thallus structure, reproduction (in brief) and economic importance of the following: *Chlorella, Cosmarium, Scenedesmus, Spirogyra*, diatoms, and *Gracilaria.*	**6hrs**
Unit V	**Fungi:**	
5.1.	Outline classification as per Alexopoulos and Mims.	**2hrs**
5.2.	Study of thallus structure, reproduction (in brief), life cycle and economic importance of the following: *Pythium, Rhizopus, Saccharomyces, Penicillium, Aspergillus, Fusarium, Agaricus.*	**7hrs**
Unit VI	**Protozoa:**	
6.	General account, structure and reproduction of-*Euglena, Paramecium*, and *Entamoeba*	**3hrs**

REFERENCES:

1. Alexopoulos, C.J., Mims, C.W. and Blackwell, M. 1996. **Introductory Mycology**, Fourth Edn. Wiley Eastern Limited, Singapore.
2. Ananthanarayanan, C. and Paniker, C.K.J. 2013. **Text Book of Microbiology**, Ninth Edn. Orient Longman Ltd., Chennai.
3. Biswas S.B. 1984. **An Introduction to Viruses**. Preface Books.
4. Brock T.D. **Biology of Microorganisms.** Prentice Hall Publications.
5. David R. Boone, Goerge M. Garrity .2012, 2011, 2009, 2005, 2001. **Bergey's Manual of Systematic Bacteriology,** 2nd Edn. Vol. I to V.Springer Publications
6. Flint S.J., Enquist L.W., Krug.2009. **Principles of Virology**, 3rd Edn. ASM Press.
7. John G. Holt .1994.**Bergey's Manual of Determinative Bacteriology,** 9th Edn. Williams and Wilkins.
8. Kotpal R.L. 2000.**Protozoa.** Rastogi Publications.
9. Vashishta, B.R., Sinha, A.K.2016. **Fungi**. S. Chand & Co.
10. Vashishta, B.R., Sinha, A.K. 2012. **Algae**. S. Chand & Co.

SEMESTER II

PRACTICAL II: MICROBIAL DIVERSITY

Experiments based on theory syllabus **14 practical (One practical of 3 hrs./wk.)**

1-2. Study of cyanobacteria- *Microcystis, Spirulina, and Anabaena*.

3-6. Study of the following algae – *Chlorella, Cosmarium, Scenedesmus, Spirogyra,* diatoms and *Gracilaria.*

7-10. Study of the following fungi – *Pythium, Rhizopus, Saccharomyces, Penicillium, Aspergillus, Fusarium, and Agaricus*.

11. Isolation of TMV in crude form (by sucrose gradient method) and inoculation to healthy plants.

12-13. Study of the following protozoa- *Euglena, Paramecium, and Entamoeba.*

14. Display of photographs/materials of – Bacteriophages, TMV, HIV, *E. coli*, *Staphylococcus aureus, Pseudomonas sp., Bacillus sp*., rickettsia, chlamydias, mycoplasmas, spirochaetes, and actinomycetes.

INTERNAL ASSESSMENT **10 marks**

Break up

Record 05 marks

Practical IA test 05 marks

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SEMESTER- III

Title: MICROBIAL GROWTH AND GENETICS (PAPER III)

Class Duration: 3 hours per week = 48 Hrs.

Marks: Theory (50) + Internal Assessment (20) = 70

Course Objectives (CO):

1. To study the nutritional requirements of microorganisms, media and methods for culturing microorganisms, cultural characteristics, factors affecting microbial growth, measurement of microbial growth.

2. To learn aseptic techniques in Microbiology **and** acquire skills in the techniques of cultivation, isolation and preservation of microorganisms.

3. To study the genetics of microorganisms and know the use of microbes as tools in the study of fundamental life processes-DNA synthesis and protein synthesis.

Course Learning outcome (CLO):

CO	After the completion of this course, the student will be able to	Cognitive Level
CO-01	To understand the nutritional, physical and chemical growth requirements of microorganisms and uptake of nutrients by microorganisms, to describe culture media	Understand
CO-02	:To master aseptic techniques and practise safety rules in the isolation of microorganisms from various sources and routine handling of pure cultures, , and to get equipped with various methods of microbial growth measurement culture-independent, methods	Analyse
CO-03	To describe the use of microorganisms in the study of fundamental life processes viz. DNA replication and protein synthesis ,elucidation of the genetic code, regulation of gene expression, etc.	Understand

	PART A **MICROBIAL GROWTH**	
Unit I	**Microbial nutrition :**	
1.1	Major nutritional types of microorganisms. Nutritional requirements of microorganisms, growth factor requirement.	**3 hrs.**
1.2	Uptake of nutrients – passive transport, facilitated, active transport, group translocation, iron uptake.	**2 hrs.**
Unit II	**Microbial Growth:**	
2.1.	Growth rate and generation time, growth curve- phases of growth and their significance. Definition of terms- Synchronous growth, Diauxic growth, Continuous growth. Quorum sensing and biofilm formation.	**2 hrs.**
2.2.	Physical and chemical factors affecting growth: temperature, radiation, pH, oxygen, carbon-di-oxide, moisture, carbon source, osmotic concentration and salinity. Microbial growth in extreme environments and its significance.	**2 hrs.**
2.3.	Measurement of growth by cell number and cell mass.	**2 hrs.**
Unit III	**Cultivation of bacteria:**	
3.1	Nutrient media – types, cultivation of aerobic and anaerobic bacteria.	**2 hrs.**
3.2	Pure cultures and cultural characteristics: Serial dilution, pure culture isolation- pour plate, spread plate, streak plate and micromanipulator technique. Colony characteristics. Non-cultivable bacteria.	**3 hrs.**
3.3	Maintenance and preservation of pure cultures- subculturing, overlaying with mineral oil, cryopreservation and lyophilization.	**2 hrs.**
	PART B **MICROBIAL GENETICS**	
Unit IV	**Introduction to Microbial Genetics**	

4.1	Historical review of genetics. Chromosomes: Prokaryotic and Eukaryotic organization. Use of microorganisms as tools in genetic studies.	**3 hrs.**
4.2	Recombination in Bacteria: Transformation, Transduction (types) and Conjugation process-F factor, Sex duction.	**3 hrs.**
4.3	Extrachromosomal genetic elements and their importance–plasmids and transposons.	**2hrs.**
Unit V	**Genetic material –structure, types and replication:**	
5.1	Chemical basis of heredity-evidence for DNA and RNA as genetic material, Watson and Crick model of DNA, DNA types. Types of RNA, structure and their functional importance.	**5 hrs.**
5.2	Replication-mode and mechanism, replication of prokaryotic DNA-general methodology	**3 hrs.**
5.3.1	Fine structure of a gene. The classical and modern concept of a gene.	**2 hrs.**
5.3.2	Gene-protein relationship: Central dogma of molecular biology. One gene-one enzyme and one gene-one polypeptide concept, colinearity of genes and proteins. Protein synthesis-transcription and translation, genetic code-features, triplet code, Wobble hypothesis, non-sense codon.	**6 hrs.**
5.4	Regulation of gene expression in prokaryotes-*lac* operon concept (positive and negative regulation).	**2 hrs.**
5.5	Introduction to bioinformatics: Definitions of bioinformatics, genomics, proteomics, metabolomics	**1 hr.**
Unit VI	**Mutations**	
6.	Nature and types, mutagenic agents: physical and chemical mutagens, damage and repair of DNA; photo reactivation and SOS repair	**3 hrs.**

REFERENCES:

1. Benjamin Lewin, Jocelyn Krebs, Elliott Goldstein .2017. **Genes XII**, 12th Edn. Prentice Hall.
2. Freifelder, D. 1984. **Essentials of Molecular Biology**. Narosa Publishing House, New Delhi.
3. Gardner, Simmons, Snustad. 2006. **Principles of Genetics**, 8th Edn. Wiley India Edn.
4. Gerhard Gottschalk. 1986. **Bacterial Metabolism,** 2nd Edn**.** Springer Publications.
5. Gupta, R.K. 1993-94. **Cytology, Genetics, Biotechnology and Biostatistics**. Rastogi Publications, Meerut.
6. Lehninger. **Biochemistry**. CBS Publishers
7. Moat, Foster, Spector.2005. **Microbial Physiology, 4th** Edn. John Wiley & Sons.
8. Nancy Trun & Janine Trempy.2004. **Fundamentals of Bacterial Genetics**. Blackwell Publishing.
9. Strickberger Monroe W. 1985. **Genetics.** Prentice Hall India.
10. Watson, Baker, Bell. 2008. **Molecular Biology of the Gene,** 6th Edn. Pearson Education.

SEMESTER - III

PRACTICAL III: MICROBIAL GROWTH AND GENETICS

Experiments based on theory syllabus 14 practical (One practical of 3 hrs. /wk.)

Expt. No.

1. Cleaning and sterilization of glassware
2. Preparation of media-nutrient broth, nutrient agar, potato dextrose agar, Czapek Dox agar, Mac Conkey's agar

3. Cultivation of microorganisms on an agar plate (point inoculation), broth, anaerobic cultivation (Candle jar/GasPak)
4. Preparation of physiological saline and serial dilution
5. Plating techniques: Spread plate, pour plate and streak plate methods.
6. Study of colony characteristics of bacteria on agar plates
7. Maintenance of stock cultures on fresh agar slants, overlaying with mineral oil, stab culture.
8. Effect of temperature on growth of microorganisms
9. Effect of pH on growth of microorganisms
10. Effect of carbon source on the growth of microorganisms
11. Effect of UV radiation source on the growth of microorganisms
12. Measurement of growth by cell mass using turbidometer / photocolorimeter / spectrophotometer
13. Measurement of growth by cell number using a haemocytometer
14. Demonstration of the following models/photographs of DNA, DNA replication, t-RNA, transcription, translation, transformation, transduction & conjugation, equipment- colony counter, pH meter, turbidometer/ photocolorimeter

INTERNAL ASSESSMENT	**10 marks**
Break up	
Record	05 marks
Practical IA test	05 marks

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SEMESTER IV
Title: MICROBIAL METABOLISM AND GENETIC ENGINEERING (PAPER IV)
Class Duration: 3 hours per week = 48 Hrs.
Marks: Theory (50) + Internal Assessment (20) = 70

Course Objectives :

1. To study the role of enzymes in metabolism

2. To study the metabolic reactions in microbes-catabolic and anabolic.

3. To learn about the tools and techniques of genetic engineering

4. To study genetic engineering in relation to Microbiology, applications and limitations of genetic engineering.

Course Learning outcome (CO):

CO	**After the completion of this course, the student will be able to**	**Cognitive Level**
CO-01	Describe the importance of enzymes in metabolism and to appreciate the biochemical unity of life.	Remember

CO-02	Understand the importance of metabolism in the growth and survival of all organisms	Understand
CO-03	Describe the relevance of Microbiology in genetic engineering	Analyze

PART A **MICROBIAL METABOLISM**		
Unit I	**Microbial enzymes**:	
1.1	Definition, nomenclature, classification, properties, cofactors and coenzymes, mode and mechanism of enzyme action, factors affecting enzyme action,	**3 hrs.**
1.2	Enzyme regulation, enzyme inhibition: competitive, non-competitive, and uncompetitive. Allosteric enzymes and their importance. Importance of microbial enzymes.	**3 hrs.**
Unit II	**Photosynthesis:**	
2.1	Definition, photosynthetic microorganisms- oxygenic and anoxygenic types, light as a source of energy, photosynthetic pigments and apparatus in prokaryotes and eukaryotes.	**3 hrs.**
2.2	Mechanism of photosynthesis in bacteria and cyanobacteria, Light reaction - Cyclic and non-cyclic photophosphorylation, Dark reaction-Calvin cycle and pyruvate synthase pathway.	**2 hrs.**
Unit III	**Respiration:**	
3.1	**Anaerobic Respiration:** Definition, breakdown of glucose to pyruvate-EMP, HMP and ED pathways. Fermentation-conversion of pyruvate to ethanol and lactic acid. Homolactic and heterolactic fermentation.	**4 hrs**
3.2.	**Aerobic Respiration:** Introduction, the ultrastructure of mitochondrion, formation of acetyl CoA from pyruvate, TCA cycle. Electron transport and oxidative phosphorylation; Chemiosmotic hypothesis. The efficiency of aerobic and anaerobic respiration as energy-yielding processes.	**4 hrs.**
Unit IV	**Nitrogen Metabolism:**	
4.1.	Biological nitrogen fixation-symbiotic and asymbiotic nitrogen fixation, nodule formation, bacteroids, *leg* haemoglobin, mechanism and biochemistry of N_2 fixation.	**3 hrs.**
4.2	Amino acid synthesis, proteolysis, nitrification and denitrification.	**2 hrs.**
PART B **GENETIC ENGINEERING**		
Unit V	**Genetic engineering-Historical perspectives and tools.**	
5.1	Genetic engineering-definition, historical perspectives and Objectives.	**2 hrs.**
5.2	Tools involved in genetic engineering:	
5.2.1	Modifying enzymes: Restriction enzymes, ligases, methylases, Reverse transcriptase	**3 hrs.**

5.2.2	Cloning vehicles- Naturally occurring plasmids; Cloning plasmids(pBR322 pUC18).Viruses as cloning vehicles (λDNA, M13, SV40).Hybrid vectors (cosmids, YAC). Cloning host (*E. coli*)	**4 hrs.**
Unit VI	**Techniques in genetic engineering:**	
6.1	Gene cloning: DNA isolation, Gel electrophoresis: Agarose gel- principle and method, transformation methods. Replica plating technique.	**5 hrs.**
6.2.1	DNA libraries: Brief account of genomic and cDNA libraries-applications.	**2 hrs.**
6.2.2	Blottings: Southern, Northern, and Western.	**1 hr.**
6.2.3	Gene screening and isolation: Molecular probes.DNA hybridization methods-colony and plaque hybridization.	**2 hrs.**
6.2.4	Restriction fragment length polymorphism (RFLP). DNA fingerprinting.	**1 hr.**
6.2.5	DNA sequencing: Brief account of Maxam and Gilbert's methods, Sanger's metho automated sequencing method.	**2 hrs.**
6.3	Applications of genetic engineering in: Agriculture, wastewater management, industries, human health. Potential problems of genetic engineering; social impact of genetic engineering.	**2 hrs.**

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SEMESTER IV

PRACTICAL IV: MICROBIAL METABOLISM AND GENETIC ENGINEERING

Experiments based on theory syllabus 14 practical (One practical of 3 hrs. /wk.)

Expt. No.

1. Study of root nodules for bacteroids.
2. Acid and gas production from carbohydrates-glucose, sucrose, lactose

3. Starch hydrolysis. 4. Gelatin hydrolysis. 5. Catalase activity. 6. Casein hydrolysis 7. Triple sugar-iron agar test 8. Nitrate reduction test 9. Isolation of DNA using coconut endosperm/onion/*E. coli*. 10-11 Replica plating technique 12. Demonstration of Bacterial Transformation using kits 13-14. Demonstration of plasmids, episomes, bacteriophages, blotting techniques, gel electrophoresis, gene cloning, colony hybridization, plaque hybridization through charts/photographs.
INTERNAL ASSESSMENT **10 marks** Break up Record 05 marks Practical IA test 05 marks

SEMESTER- V
Title: ENVIRONMENTAL MICROBIOLOGY (PAPER V)
Class Duration: 3 hours per week = 48 Hrs.
Marks: Theory (70) + Internal Assessment (30) = 100

Course Objectives :

1. To study the diversity of microorganisms in different parts of the environment, techniques used to isolate and study them.

2. To study of the role and significance of microorganisms in the environment

3. To study of the methods used to harness the metabolic potential of microorganisms to address problems facing the society like pollution, energy production, etc.

Course Outcome:

CO	**After the completion of this course, the student will be able to**	**Cognitive Level**
CO-01	To understand the uniqueness of microbial life in ecosystems	Understand
CO-02	To describe the role of microorganisms in different parts of the environment	Remember
CO-03	**To** understand the importance of microbial processes in environmental sustainability.	Apply

Unit I	**Microbiology of air:**	
1.1	Introduction: Definition, history and development, aim and scope of aerobiology.	**2 hrs.**
1.2	Microbes and atmosphere: Atmospheric layers, sources of microorganisms, air spora of indoor and outdoor environment, factors affecting air spora, significance of airborne microbes, and management of air-borne microbes.	**7 hrs.**
1.3	Techniques of trapping air-borne microorganisms: Gravity slide, Petri plate exposure, Vertical cylinder spore trap, Hirst spore trap, rotorod sampler, Andersen sampler, impinger and filtration. Advantages and disadvantages of the techniques.	**6 hrs.**
Unit II	**Microbiology of water:**	
2.1	Introduction: Natural waters, distribution of microorganisms in the aquatic environment.	**3 hrs.**
2.2.1	Water pollution: Sources of water pollution.	**2 hrs.**
2.2.2	Water-borne pathogens- Viral (Hepatitis A virus, Polio virus), bacterial (*S. Typhi, S. paratyphi*, species of *Shigella* and *V. cholerae*), and protozoal (*E. histolytica, Giardia lamblia*). Eutrophication of lakes. Biological indicators of water pollution	**5 hrs.**

2.3	Determination of sanitary quality of water: Standard analysis of water quality- Presumptive, confirmed and completed the test, determination of MPN, SPC, H_2S strip test, membrane filter technique, biochemical tests for coliforms- IMViC reactions.	**6 hrs.**
2.4	Water purification in municipal water supplies.	**2 hrs.**
Unit III	**Microbiology of wastewater:**	
3.1	Introduction: Sources of wastewater- domestic, agricultural and industrial. Physical, chemical and microbiological characteristics of wastewater.	**3 hrs.**
3.2.1	Wastewater treatment: Single dwelling unit-septic tank, Imhoff tank; Municipal wastewater treatment-primary (screening, coagulation and sedimentation), secondary (trickling filter, activated sludge process, oxidation pond), tertiary (disinfection, reverse osmosis, ion exchange method and dialysis in brief) and reclamation of wastewater.	**7 hrs.**
3.2.2.	Solid waste recycling: Composting and anaerobic digestion process- biogas plant.	**3 hrs.**
3.3.	Bioremediation: Biodegradation of pesticides and other pollutants (oil spills).	**2 hrs.**

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SEMESTER – V

PRACTICAL V: ENVIRONMENTAL MICROBIOLOGY

Experiments based on theory paper V 14 practical (One practical of 3 hrs./wk.)

1. Isolation of air borne microorganisms (bacteria and fungi) by gravity slide method and Petri plate exposure method.

2-3. Demonstration of air sampling equipment /photographs of vertical cylinder spore trap, rotorod sampler, Hirst spore trap, Andersen sampler, liquid impingement method (bead bubbler), membrane filter

4.Microscopic observation of different water samples for microorganisms as biological indicators of water pollution.

5 -6. Standard analysis of water samples

7. Determination of MPN

8. IMViC reactions

9. Water quality testing by H_2S strip test

10. Display of photographs of water purification process (baffles, flocculator, clarifier, sand filter, back wash, chlorinometer and chloroscope

11. Determination of BOD of sewage

12. Estimation of total solids of sewage

13. Display of photographs **of** waste water treatment process **(**septic tank, Imhoff tank, trickling filter, activated sludge process, oxidation pond, sedimentation tank, anaerobic digester/biogas plant).

14. Visit to water treatment plant/ sewage treatment plant/ industrial effluent treatment plant.

INTERNAL ASSESSMENT: 15 MARKS

Breakup

Record	:	5 marks
Report	:	5 marks
Practical IA test	:	5 marks

SEMESTER- V
MICROBIOLOGY
Title: SOIL AND AGRICULTURAL MICROBIOLOGY (PAPER VI)
Class Duration: 3 Hours per week = 48 Hrs.
Marks: Theory (70) + Internal Assessment (30) = 100

Course Objectives :

1. To isolate and study microorganisms in soil, their role in soil processes like biogeochemical cycling of elements, decomposition of dead plant and animal remains, etc.

2. To study the interactions within microbial communities and between microorganisms and plants and animals.

3. To study plant pathogenic microorganisms, plant growth-promoting(biofertilizers) and plant-protecting microorganisms(biopesticides)

Course learning outcome (CO):

CO	**After the completion of this course, the student will be able to**	**Cognitive Level**
CO-01	To describe the microbial diversity in soil and the ecological importance of microbial processes	Understand
CO-02	To evaluate the importance of microorganisms in agriculture – beneficial and harmful aspects.	Remember
CO-03	To identify symptoms and causative agents of plant diseases and suggest control measures	Apply

Unit I	**Microbiology of Soil:**	
1.1.	Introduction: Brief history of Soil Microbiology. Soil-Definition, types. Soil profile and physical characteristics of soil -mineral particles, organic residues, water and gases.	**3 hrs.**
1.2.	Microbial flora of soil: A brief account of bacteria, fungi, algae, actinomycetes, protozoa and viruses. A brief account of factors affecting soil microflora.	**3 hrs.**
1.3.	Role of microbes in soil processes: Biogeochemical cycles-Carbon and Nitrogen cycles; biodegradation of pectin, cellulose and lignin.	**5 hrs.**
1.4.	Plant – microbe interaction: Rhizosphere microorganisms- Rhizosphere and rhizoplane, Plant growth -promoting bacteria (PGPR), Mycorrhizae.	**4 hrs.**
Unit II	**Microorganisms in Agriculture:**	
2.1.1.	Biofertilizers: Definition, types- Nitrogen- fixing, phosphate- solubilizing and cellulolytic microbes, mass production of bacterial inoculants	**6 hrs.**

	(*Rhizobium*, *Azotobacter*, *Azospirillum*, cyanobacteria), modes of application, advantages and limitations.	
2.1.2.	Biopesticides: Definition, types- bacterial, viral, fungal and protozoal, mode of action, factors influencing, genes involved, target pests, use of transgenic plants. Microbial herbicides.	**8 hrs.**
2.2.1.	Interactions among microorganisms-neutral (neutralism), positive (mutualism, commensalism), negative (competition, antagonism, predation and parasitism- mycoparasitism, nematophagy).	
2.2.2.	Biological control-Introduction, role of negative interactions among microorganisms in biological control. Applications of biological control on field	
Unit III	**Plant pathology**:	
3.1.	Introduction: Historical developments in brief, classification of plant diseases, principles of infection and spread of diseases in general. Defense mechanism in plants- structural and biochemical.	**5 hrs.**
3.2.	Plant diseases: Study of symptoms, etiology, epidemiology and management of the following diseases-Bean mosaic, Sandal spike, citrus canker, downy mildew of grapes, powdery mildew of mulberry, rust of sorghum, blast disease of paddy, tikka disease of groundnut. Brief account of post – harvest pathology.	**9 hrs.**

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SEMESTER V

PRACTICAL VI: SOIL AND AGRICULTURAL MICROBIOLOGY

Experiments based on theory paper VI **14 practical (One practical of 3 hrs./wk.)**

Expt. No.

1. Isolation and identification of fungi from soil by serial dilution method.
2. Isolation and enumeration of bacteria from soil by serial dilution method.
3. Study of VAM
4. Biofertilizer samples.
5. Isolation of *Rhizobium* from root nodules
6. Study of *Anabaena* from *Azolla.*
7. Cultivation of *Azolla*.
8. Study of antagonism between microorganisms.
9. Study of soil algae.
10. Isolation and identification of soil protozoa.
11-13. Study of plant diseases: Bean mosaic, Sandal spike, Gram staining of citrus canker specimen and mounting of fungal specimens - downy mildew of grapes, powdery mildew of mulberry, rust of sorghum, blast disease of paddy and tikka disease of groundnut)
14. Visit to agricultural research station. **Every student shall compulsorily submit an independent report on the visit along with practical record for internal assessment.**

INTERNAL ASSESSMENT: 15 MARKS

Breakup

Record	:	5 marks
Report	:	5 marks
Practical IA test	:	5 marks

SEMESTER- VI
MICROBIOLOGY
Title: FOOD MICROBIOLOGY AND INDUSTRIAL MICROBIOLOGY (PAPER VII)
Class Duration: 3 Hours a week = 48Hrs
Marks: Theory (70) + Internal Assessment (30) = 100

Course Objectives :

1: To study the applied branches of Microbiology -food, dairy and industrial microbiology, applications of microorganisms in these fields.

2: To study the use of microorganisms in food production, processing, food preservation, role of microbes in food spoilage and food-borne infections and intoxications.

3. To learn about food regulation standards and laws.

Course learning outcome (CO):

CO	After the completion of this course, the student will be able to	Cognitive Level
CO-01	Describe microorganisms important in food, dairy and industry and develop experimental skills for testing milk and different foods for the presence of microorganisms	Understand
CO-02	Describe methods to prevent spoilage of food and dairy products	Remember
CO-03	Understand the applications of microorganisms to produce useful products	Apply

PART A FOOD MICROBIOLOGY		
Unit I		
1.1.	Introduction to food microbiology: Definition, concepts and scope.	**2 hrs.**
1.2.1.	Microbes and food: Food as a substrate for growth of microorganisms (Factors that favour or inhibit the growth of microorganisms).	**3 hrs.**
1.2.2.	Sources of contamination, principles of spoilage. Microbial spoilage of food – Fruit, vegetables, meat, poultry, canned food.	**3 hrs.**
1.2.3.	Fermented foods- Idli, Sauerkraut	**1 hr.**
1.3.1.	Principles and methods of food preservation: Physical methods -High temperature (canning), low temperature, drying (solar drying, drum drying, spray drying), radiation, Modified Atmosphere Storage	**5 hrs.**

	and Modified Atmosphere Packaging (MAP).	
1.3.2.	Chemical methods – Chemical preservatives-Salient features of the chemical preservatives (propionates, benzoates, sorbates, nitrates and nitrites, sulfur dioxide and sulphites, sugars and salt). Bacteriocins as preservatives.	**2 hrs**.
Unit II	**Microbial toxins and food safety**	
2.1.	Food- borne infections and intoxications: Bacterial intoxication – Botulism, Staphylococcal intoxication. Bacterial infection - Salmonellosis Mycotoxins: Types and importance of toxins with reference to aflatoxins.	**4 hrs.**
2.2.	Food safety and quality control (microbiological criteria and microbiological quality control). A brief account of food laws and standards.	**2 hrs.**
Unit III	**Dairy Microbiology:**	
3.1.	Introduction to the microbiology of milk. Sources of milk contamination. Methods to detect microbial spoilage -SPC, Reductase test, Phosphatase test, clot on boiling test (COB).Biochemical changes in milk-souring, gassy fermentation, proteolysis, lipolysis and ropiness.	**4 hrs.**
3.2.	Fermented dairy products: A brief account of characteristics and therapeutic value. Starter culture and its role. **Probiotics**- Acidophilus milk, yoghurt. Cheese - Preparation and types.	**3 hrs.**
3.3.	Preservation of milk and milk products – Pasteurization, Sterilization.	**1 hr.**

PART B **INDUSTRIAL MICROBIOLOGY**		
Unit IV		
4.1.	Introduction, definition and scope of industrial microbiology. Microorganisms of industrial importance Types of stock culture. Strain improvement.	**3 hrs.**
4.2.	Fermentation media: A brief account of – Production medium, inoculum medium; Raw materials – Molasses and types, corn steep liquor, sulphite waste liquor and whey. Buffers, precursors, inhibitors and antifoam agents.	**2 hrs.**
4.3.	Design of a typical fermentor: Devices for aeration, agitation	**1 hr.**
4.4.	Fermentation processes – surface, submerged and solid state fermentation; Types- Batch and Continuous fermentation.	**2 hrs.**
4.5.	Downstream processing – Precipitation, filtration, centrifugation, distillation, cell disruption, solvent recovery, drying, crystallization	**2 hrs.**
Unit V		
5.1.	Industrial production and uses of the following : Ethyl alcohol, wine**;** Antibiotic – Penicillin**;** Organic acid – lactic acid**;** Enzyme – amylase; Single Cell protein (Salient features and nutritional value) – *Spirulina, Chlorella* Mushroom cultivation – Oyster mushroom (bag method), nutritional value.	**7 hrs.**
5.2.	Role of microorganisms in production and recovery of minerals and petroleum.	**1 hr.**
REFERENCES:		

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SEMESTER VI

PRACTICAL VII: FOOD MICROBIOLOGY & INDUSTRIAL MICROBIOLOGY

Experiments based on theory paper VII 14 practical (One practical of 2 hrs./wk.)

1. Isolation and enumeration of bacteria from food utensils
2. Isolation and identification of fungi from food utensils.
3. **a.** Isolation and enumeration of bacteria from spoiled vegetable
   b. Isolation and identification of fungi from spoiled vegetables.
4. **a.** Isolation and enumeration of bacteria from spoiled fruit
   b. Isolation and identification of fungi from spoiled fruit
5. Isolation and identification of Aspergillus on groundnut by blotter method.
6. Microscopic examination of idli batter.
7. Quantitative examination of bacteria in raw and pasteurized milk by SPC method.
8. **a**. Phosphatase test and COB test
   b. Turbidity test to detect boiled and unboiled (raw milk)
9. Methylene blue reductase test to determine the quality of milk.
10. **a**. Preparation of wine from grapes.
    b. Preparation of alcohol using jaggery /molasses.
11. **a**. Oyster Mushroom cultivation: Bag method
    b. Culturing of *Spirulina /Chlorella*
12. Production and estimation of citric acid from fungi
13. Display of food, dairy and industrial products-cheese, yogurt, bread, molasses and wine
14. Visit to food industry/ food research laboratory, dairy industry, distillery.

INTERNAL ASSESSMENT: 15 MARKS

Breakup

Record : 5 marks
Report : 5 marks
Practical IA test : 5 marks

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SEMESTER- VI
MICROBIOLOGY
Title: IMMUNOLOGY AND MEDICAL MICROBIOLOGY (PAPER VIII)
Class Duration: 3 Hours per week = 48 Hrs.
Marks: Theory (70) + Internal Assessment (30) = 100

Course Objectives:

1: To study of the structure and function of the immune system, and its response towards foreign and self –antigens.

2. To study the nature of antigen and antibody; role of antigen-antibody reactions in immunity, autoimmune disorders, hypersensitivity.

3. To learn the applications of antigen-antibody reactions in sero-diagnosis of diseases.

4. To understand the application of Immunology in prophylaxis of diseases using vaccines.

5. To study of the role of microorganisms in human health, concepts of infection, pathogenicity, pathogenesis, virulence, some examples of pathogenic microbes and the diseases they cause, treatment of diseases using antibiotics, antibiotic resistance development.

Course learning outcome (CO):

CO	After the completion of this course, the student will be able to	Cognitive Level
CO-01	understand the structure and function of the immune system in immunity, hypersensitivity and autoimmune diseases to describe immunoprophylaxis of diseases using vaccines	Understand
CO-02	understand the importance of microorganisms in human health	Understand
CO-03	identify symptoms, etiology of common diseases, describe preventive and treatment methods	Apply

PART A IMMUNOLOGY		
Unit I		
1.1.	Historical account of immunology.	**2 hrs.**
1.2.	Types of immunity – Innate (non–specific) and Adaptive immunity (specific). Antibody (humoral)-mediated immunity and cell-mediated immunity.	**2 hrs.**
1.3.	Introduction to the immune system. Cells and tissues of immune system – Structure and role of primary lymphoid organs (bone marrow, thymus), secondary lymphoid organs (spleen, lymph nodes and tonsils). B and T lymphocytes, phagocytes, killer cells, NK cells. Hypersensitivity (in brief).	**6 hrs.**
1.4.	Microbial mechanisms for escaping host defences - Evading the Complement System, Resisting Phagocytosis, and Survival inside Phagocytic cells, evading the specific immune response.	**2 hrs.**

Unit II		
2.1.	Antigens – nature and types. Antibodies – structure of IgG. Classes of antibodies and their functional diversity.	**4 hrs.**
2.2.	Antigen-antibody reactions – salient features. Agglutination reaction- Widal test, Precipitation reaction- VDRL test, Neutralization test, Opsonization. Immunotechniques – RIA, ELISA, ODD.	**4 hrs.**
2.3.	Human blood types and Rh factor. Complement system (in brief) – complement fixation test.	**2 hrs.**
Unit III		
3.	Immunoprophylaxis – Vaccine – Types –Killed, live attenuated (bacterial and viral), toxoid, sub-unit vaccines, recombinant vaccine with an example each. National immunization schedule (tabular form).	**2 hrs.**
PART B MEDICAL MICROBIOLOGY		
Unit IV		
4.1.	History and development of Medical Microbiology. Normal microflora of the human body.	**3 hrs.**
4.2.	Infection – types of infection, modes of transmission, and portals of entry.Pathogenesis, Pathogenicity – Virulence – attenuation and exaltation.	**5 hrs.**
4.3.1.	Medically important microorganisms. Pathogen – morphology, cultural and biochemical characteristics, classification, resistance and pathogenesis, clinical symptoms, laboratory diagnosis, epidemiology, prophylaxis and treatment of the following: Bacterial diseases – tuberculosis, syphilis.	**3 hrs.**
4.3.2.	Viral diseases – Hepatitis B, AIDS.	**2 hrs.**
4.3.3.	Fungal diseases- Candidiasis, Dermatomycosis (*Tinea* – ringworm infection).	**2 hrs.**
4.3.4.	Protozoal diseases – Malaria, Trichomoniasis	**2 hrs.**
4.3.5.	Emerging diseases	**2 hrs.**
Unit V		
5.	Chemotherapy: General characteristics and types of antibiotics. Mode of action of penicillin, streptomycin and chloramphenicol. Antibiotic resistance- Multidrug resistance in brief.	**5 hrs.**

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SEMESTER VI

PRACTICAL VIII: IMMUNOLOGY AND MEDICAL MICROBIOLOGY

Experiments based on theory paper VIII 14 Practical (One practical of 2 hrs./wk.)

1. Determination of blood group and Rh factor.
2. Demonstration of precipitation reaction –Double diffusion in two dimensions (Ouchterlony procedure).
3. Culturing of microorganisms from blood sample
4. Culturing of microorganisms from urine sample
5. Microbial flora of human skin
6. Microbial flora of oral cavity (tooth and mouth)
7. Antibiotic sensitivity test.
8. Determination of MIC of antibiotics
9. Demonstration of VDRL test using kits.
10. Demonstration of Widal test using kits.

11-13. Microscopic observation/ display of photographs of human pathogens- *Mycobacterium tuberculosis*, *Treponema pallidum*, Hepatitis B virus, HIV, *Candida albicans*, *Tinea*, Plasmodium, *Trichomonas vaginalis*.

14. Visit to pharmaceutical and pathological laboratories, a report of which shall be submitted along with the practical record for internal assessment.

INTERNAL ASSESSMENT: 15 MARKS

Breakup

Record	:	5 marks
Report	:	5 marks
Practical IA test	:	5 marks

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DSE
Paper code: 18MBSC-A1
Title of the Paper: ANALYTICAL METHODS IN MICROBIOLOGY
Teaching hours- 32(2 hrs./week)
No. of credits- 2

Course Objectives:

1.Students are expected to gain a fundamental and practical understanding of the use of analytical instrumentation in microbiological research and applications.

Course Description: The course combines lectures and laboratory experiments with hands-on training in the use of analytical instrumentation and experimental design in microbiological research and applications. Analytical methods that are introduced include gas chromatography, GC-mass spectrometry, liquid chromatography (HPLC), ion chromatography, and select molecular methods like gel electrophoresis, genome sequencing, PCR. The course serves as an introduction to chromatographic analysis, basic principles of mass spectrometry, and reviews different choices of methods and instruments. This includes familiarization with different analytical software and methods of data analysis and interpretation. The course requires collaboration with local research institutes like CFTRI, DFRL and University Life Science departments for training and exposure to their instrumentation facilities.

Course learning outcome (CO):

CO	After the completion of this course, the student will be able to	Cognitive Level
CO-01	Demonstrate an understanding of the principles of instrumental analysis.	Apply
CO-02	Show the ability to interpret data from GC, GC-MS and HPLC applications.	Remember
CO-03	Apply the scientific method to collect, interpret, and present scientific data in microbiology and related fields	Apply

Examinations and Assignments: Internal assessment (based on laboratory performance and participation, written reports, presentations) for 20 marks and final examination for 30 marks.

SYLLABUS:

Unit I		
1.1.	**Antibiotic sensitivity tests**: Diffusion Methods. Zone of inhibition. Dilution Methods of Antibiotic Assays - Streak-Plate Method, Serial Dilution Assay, Antiluminescent Assay, Photometric assay. Microbiological Assay of Penicillin.	**2 hrs.**
1.2.	**Microbiological Assay of vitamins**: Use of auxotrophs. Types of assay- Large Plate Methods-Microbiological Assay of Riboflavin.	**2 hrs.**
Unit II		

2.1.	**Chromatography:** Principles and applications of paper chromatography (including Descending and 2-D), Thin layer chromatography. Separation of mixtures by paper / thin layer chromatography. Column packing and fraction collection. Gel filtration chromatography, ion-exchange chromatography and affinity chromatography, GLC, HPLC. Chromatographic separation of protein mixtures	**5 hrs**.
2.2.	**Centrifugation:** Preparative and analytical centrifugation, fixed angle and swinging bucket rotors. RCF and sedimentation coefficient, principle and application of differential centrifugation, density gradient centrifugation and ultracentrifugation. Separation of components of a given mixture using a laboratory-scale centrifuge	**5 hrs.**
2.3.1.	**Spectrophotometry:** Introduction to IR and NMR and their applications.Applications of Radioisotopes in biosciences. Atomic spectroscopy: Principles and application of Atomic Absorption/Emission Spectrometer. GC-MS.	**3 hrs.**
2.3.2.	Principle and use of absorption spectra of biomolecules. Analysis of biomolecules using UV and visible range. Determination of λ max for an unknown sample and calculation of extinction coefficient.	**3 hrs.**
2.3.3.	Colourimetry and turbidometry.	**2 hrs.**
Unit III		
3.1.	**Electrophoresis**: Principle and applications of native polyacrylamide gel electrophoresis, SDS- polyacrylamide gel electrophoresis, 2D gel electrophoresis, Isoelectric focusing, and Agarose gel electrophoresis. Separation of protein mixtures by Polyacrylamide Gel Electrophoresis (PAGE).	**4 hrs.**
3.2.	**Modern analytical techniques and biosensor technology**	
3.2.1.	DNA sequencing: Principles and Methods, Automated DNA sequence Analyzer Blotting techniques and FISH RFLP, RAPD, VNTR, STR and SNP analysis Chemical synthesis of DNA PCR Technology: Principle, Methods and Applications Introduction to Biosensor Technology	**6 hrs.**

Reference Books:

1. Cooper, G.M. and Hausman, R.E. 2009. **The Cell: A Molecular Approach**. 5th Edn. ASM Press & Sunderland, Washington D.C., Sinauer Associates, MA.
2. De Robertis, E.D.P. and De Robertis, E.M.F. 2006. **Cell and Molecular Biology**, 8th Edn. Lipincott Williams and Wilkins, Philadelphia.
3. Karp, G. 2010. **Cell and Molecular Biology: Concepts and Experiments,** 6th Edn. John Wiley & Sons. Inc.
4. Nelson, D.L. and Cox, M.M. 2008**. Lehninger Principles of Biochemistry,** 5th Edn. W.H. Freeman and Company.
5. Nigam, A. and Ayyagari, A. 2007. **Lab Manual in Biochemistry, Immunology and Biotechnology**. Tata McGraw Hill.
6. Willey, M.J., Sherwood, L.M. &Woolverton, C. J. 2013. Prescott, Harley and Klein's

Microbiology. 9th Edn. McGraw Hill.
7. Wilson, K. and Walker, J. 2010. **Principles and Techniques of Biochemistry and Molecular Biology,** 7th Edn. Cambridge University Press.

Paper code: 18MBSC-A2
Title of the Paper: APPLIED MYCOLOGY
Format: 2 hrs. per week lecture &/ laboratory period
Teaching hours- 32(2 hrs. /week)
No. of credits- 2

Course Objectives:
1. To widen the knowledge on fungi.
2. To develop the practical skills required for safe handling and study of fungi.
3. To appreciate the importance and applications of fungi
4. To provide a thorough understanding of the ecology and physiology of yeasts and moulds in key food chains and the methods for detecting and controlling contamination, mycotoxin contamination, and use of hurdle technology for improving shelf life.

Course description:
The course combines lectures and laboratory experiments on isolation, culturing, microscopic observation of fungal morphology and identification, physiology, genetics, diversity, ecology, economic importance and applications in industry, agriculture and medicine.

Course Learning outcomes(CO)

CO	After the completion of this course, the student will be able to	Cognitive Level
CO-01	Isolate and identify commonly isolated fungi up to genus level with the aid of illustrated manuals. Demonstrate a detailed knowledge and recognition of the key groups of fungi responsible for spoilage fungi in key food chains.	Apply
CO-02	Appreciate fungi as the most diverse group of microorganisms	Understand
CO-03	Appreciate the enormous importance and applications of fungi.	Apply

SYLLABUS:

Unit I

1. **Introduction:** Diversity, taxonomy and economic importance of fungi **3 hrs.**

Unit II

2.1. **Food mycology:** **3 hrs**.
Analysis and rapid diagnostics of fungal food spoilage – traditional vs. modern approaches. Detection of Mycotoxins in foods**.**

2.2. Factory environment –mold problems. Modified atmosphere packaging and hurdle technology to control spoilage molds. **2 hrs**.

2.3. Application of fungi in food industry: Flavor & texture, Fermentation, Baking, Organic acids, Enzymes, Mycoproteins, Secondary metabolites - Industrial production of Penicillin, Citric Acid and Alcohol. Edible Mushrooms: Medicinal and Nutritional Value, Mushroom Cultivation. **6 hrs**.

Unit III

3. **Role of fungi in biotechnology:** Scope and Techniques of Fungal Biotechnology. Fungi in Relation to Pollution, Fungi in Bioremediation. **3 hrs.**

Unit IV

4.1. **Role of fungi in Agriculture** -Biofertilizers, Mycorrhizae. Biological control - Fungi as Biopesticides- (Mycofungicides, Mycoherbicides, Mycoinsecticides, Myconematicides). *Trichoderma* as a biological control agent. **4 hrs.**

4.2. **Fungi as Pathogens:**

4.2.1. **Plant pathogenic fungi:** Etiological agents of Late Blight of Potato, Downy Mildew of Bajra, Powdery mildew of cucurbits, Blight of paddy, Tikka disease of groundnut, Head smut of Sorghum. **6 hrs.**

4.2.2. **Medical mycology**: Classification of Medically important fungi. The general identification process of medically important fungi. Study of aetiology, lab diagnosis, pathogenesis and treatment of Superficial (*Trichophyton*) and Systemic (Candidiasis) diseases of human beings. **5 hrs.**

Reference Books:

1. Agrios, G.N. 1997. **Plant Pathology**, 4th Edn. Academic Press, U.K.
2. Alexopoulos, C.J., Mims, C.W., Blackwell, M. 1996. **Introductory Mycology**, 4th Edn. John Wiley & Sons (Asia), Singapore.
3. Deacon, J. W.1997. **Modern Mycology,** 3rd Edn, Blackwell Science Publishers, London.
4. Mueller, G. M., Bills, G.F. and Foster, M.S. 2004. **Biodiversity of Fungi**, Elsevier Academic Press, New York.
5. Sethi, I.K. and Walia, S.K. 2011. **Text book of Fungi and Their Allies**. Macmillan Publishers India Ltd.
6. Sharma, P.D. 2011. **Plant Pathology.**Rastogi Publication, Meerut, India.
7. Webster, J. and Weber, R. 2007. **Introduction to Fungi**, 3rd Edn. Cambridge University Press, Cambridge.
8. Ranganathan, S., Gokul Shankar, S. and Ranjith, M.S. 2001. **Fungal Diseases and Diagnosis,**

Vol. I, 1st Edn., Tamil Nadu State Council for Science and Technology, Chennai.

Paper code: 18MBSC-A3
Title: BASICS OF RESEARCH METHODOLOGY
Format: 2 hrs per week lecture &/ laboratory period
Teaching hours- 32 (2 hrs./week)
No. of credits- 2

Course Objectives:

1. To gain an overview of research intent and design, methodology and technique, format and presentation, and data management and analysis informed by commonly used statistical methods.
2. To understand the terminologies in research

Course Description:

The course includes lectures on topics related to microbiological research, library work and searching for literature in databases.

Course Learning outcome(CO)

CO	After the completion of this course, the student will be able to	Cognitive Level
CO-01	Understand some basic concepts of research and its methodologies	Understand
CO-02	Identify, explain, and apply the basic concepts of research, such as variables, sampling, reliability, and validity.	Apply
CO-03	Recognize ethical issues in research	Analyse
CO-04	Select and define appropriate research problem and parameters	Remember
Co-05	Prepare a project proposal (to undertake a project	Create
CO-06	Organize and conduct research (advanced project) in a more appropriate manner. and write a research report and dissertation	Apply

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SYLLABUS

Unit- I		
1.1.	**Introduction:** Definition of Research, essential qualities of research-validity and reliability. Research and professions.	**2 hrs.**
1.2.	**Research process**- Defining the research problem and title, objectives and scope of the study, a period for the study. Initial literature review- sources (Library catalogue. Electronic Databases, Journals and Newspapers), setting and fixing hypotheses.	**6 hrs**.
1.3.	Research design and strategy- basic elements of research design, types of research design.	**2 hrs.**

1.4.	Research methods – Qualitative versus quantitative techniques, sample and sampling procedures, Data collection methods -primary and secondary sources of data, Analysis of collected data- processing of data, editing, classification, tabulation, interpretation and reporting findings.	**6 hrs.**
Unit- II		
2.1.	**Basic Statistics for Research** -Sample and population, Central Tendency- Mean, Median, Mode Summation notations, Standard Deviation, Standard error, Variation, Q-test, T-test, F-test, R-test.	**4 hrs.**
2.2.	Test of hypotheses, the null hypothesis and alternative hypothesis, chi-square test.	**3 hrs.**
Unit-		
3.1.	Ethical considerations, research limitations	**1 hr.**
3.2.	Originality of research. Conflicts of interest, Copyright and patent law, Plagiarism.	**2 hrs.**
Unit-		
4.1.	Professional Science Writing -Research proposal and report writing.	**3 hrs.**
4.2.	Principles Of presentation- Data presentation-Dot diagram, bar diagram, Histogram, frequency curve. Communication- oral, posters. Abstract and manuscript preparation.	**3 hrs.**

Reference Books:

1. Ann Bowling, A. and Ebrahim S. 2005. **Handbook of Health Research Methods**. Open University Press, Two Penn Plaza, New York, NY.
2. Baumgartner, T. and Hensley, L. 2006. **Conducting and Reading Research in Health and Human Performance** 4th Edn. McGraw-Hill, New York.
3. Chilisa, B.2011. **Indigenous Research Methodologies**.SAGE Publications, London.
4. Flick, U. 2011. **Introducing Research Methodology**: A Beginner's Guide to Doing a Research Project. SAGE Publications, London
5. Kumar, R., 2010.**Research Methodology: A Step-by-Step Guide for Beginners.** 3rd Edn. SAGE Publications, London
6. Nicholas Walliman.2011.**Research methods- the basics**. Taylor & Francis e-Library.

Paper code: 18MBSC-A4
Title: DIAGNOSTIC VIROLOGY
Format: 2 hrs per week lecture &/ laboratory period
Teaching hours- 32(2 hrs. /week)
No. of credits- 2

Course Objectives:

1. To understand the diagnostic tools for viral infections.
2. Collection, transportation and processing of clinical samples from infected patients.

Course Description:

The paper focuses on the essentials of clinical lab skills for Microbiology. The course covers the area of Clinical Virology and Microbiology diagnostics including serodiagnostics and molecular diagnostics. Coursework may include lectures and training in a lab or hospital, as part of the

curriculum, where the students have the opportunity to use state-of-the-art instrumentation. They will also meet for a few hours for reading, discussing and analyzing scientific articles from Clinical Virology journals as a partial requirement for internal assessment.

Learning outcome:

CO	After the completion of this course, the student will be able to	Cognitive Level
CO-01	Describe and demonstrate the proper safety precautions while handling and disposing of infectious material including those potentially containing organisms like HIV or Hepatitis virus	Apply
CO-02	Perform various techniques used in the proper collection, transportation, and processing, microbiological analysis of the clinical specimens and accurate interpretation of results.	Analyse
CO-03	Have a working knowledge of the principles of techniques performed	Understand
CO-04	pursue courses which lead them to careers such as clinical laboratory technologists, Clinical microbiologist, Research technician, Lab research analyst, etc. in public and private hospitals, laboratories and physicians' clinics	Apply
CO-05	Appreciate the role of microbiology in relation to the professions in the Clinical laboratory set up and everyday life	Apply

SYLLABUS:

Unit I		
1.1.	History of Virology and introduction to Diagnostic Virology**.**	**2 hrs.**
1.2.	Collection of clinical samples (Sputum, Skin, Blood, Urine and Stool) with proper precautions. Method of transport of clinical samples to the laboratory and storage, safe handling, transport.	**3 hrs.**
Unit II		
2.1.	Classification of animal viruses. Modes of transmission of viral diseases. Viral Zoonotic infections.	**3 hrs.**
2.2.	Isolation, Identification, Cultivation and, Purification of plant and animal viruses	**2 hrs.**
Unit III		
3.1.	Development of diagnostic virology and laboratory diagnosis of viral diseases. Laboratory management and biosafety (biological containment).	**5 hrs.**
3.2.1.	Methods for the diagnosis of herpesviruses, adenoviruses, rotaviruses, coronaviruses, and viruses associated with rash diseases, hepatitis viruses, enteroviruses, retroviruses, poxviruses, orthomyxoviruses, paramyxoviruses and arboviruses. Newly emerging viral diseases.	**12 hrs**.
3.2.2.	Conventional rapid molecular detection and serodiagnosis of viruses. - Identification of infectious agents through ELISA & PCR. Interpretation of Lab investigation.	**4 hrs.**
3.3.	Antiviral chemotherapy. Viral vaccines. Interferons.	**1 hr.**

Reference Books

1. Bachmann, P.A.2012. **New Developments in Diagnostic Virology**. Springer-Verlag New York,

LLC.
2. Jane Flint, S., Enquist, L.W., Vincent, R.2015. **Principles of Virology**. ASM Press.
3. John, R. S. and Warnes, A. **Diagnostic Virology Protocol: Method in Molecular Biology**. Humana Press.
4. Kudesia, G.2009. **Clinical and Diagnostic Virology**. Cambridge University Press.
5. Leonard, C. N.2009. **Virology: Molecular Biology and Pathogenesis**. ASM Press.
6. Stephenson, J.R., and Warnes, A. 2011. **Diagnostic Virology Protocols.** Springer-Verlag New York, LLC.
7. Wagner, E.K., Hewlett, M.J. and Bloom, D.C **Basic Virology**, 3rd Edn.2009. Blackwell Publishing.

B.Sc. Microbiology CBCS Scheme, 2018 onwards
Paper code: 18MBSC-B1
Title: CLINICAL BACTERIOLOGY
Format: 2 hrs per week lecture &/ laboratory period
Teaching hours- 32(2 hrs. /week)
No. of credits- 2

Course Objectives:
1.To develop procedural skills in Clinical Microbiology along with the thinking processes that relate the following factors to each other.
1. Patient characteristics
2. Types of infections
3. Specimen requirements
4. Microscopic examinations
5. Culture media: growth characteristics and amounts
6. Specific organisms and their identification techniques.
7. Treatment and procedures relating to that treatment.

Course Description:
The paper focuses on the essentials of clinical lab skills for Microbiology. The course covers the area of Clinical Bacteriology and Microbiology diagnostics including serodiagnostics and molecular diagnostics, antibiotic therapy. Coursework may include lectures and training in a lab or hospital, as part of the curriculum, where the students have the opportunity to use state-of-the-art instrumentation. They will also meet for a few hours for reading, discussing and analyzing scientific articles from Clinical Microbiology journals as a partial requirement for internal assessment.

Course Learning Outcome (CO)

CO	After the completion of this course, the student will be able to	Cognitive Level
CO-01	Describe and perform various techniques used in the proper collection, transportation, and processing, microbiological analysis of the clinical specimens and accurate interpretation of results.	**Apply**
CO-02	Have a working knowledge of the principles of techniques performed	**Apply**

CO-03	Get prepared to pursue courses which lead them to careers such as clinical laboratory technologists, Clinical microbiologist, Research technician, Lab research analyst, etc.in public and private hospitals, laboratories and physicians' clinics.	**Apply**
CO-04	Appreciate the role of microbiology in relation to the professions in the Clinical laboratory set up and everyday life	**Understand**

SYLLABUS:

Unit I		
1.1.	**Introduction to Clinical Bacteriology:** Historical development of Bacteriology, classification of pathogenic bacteria.	**2 hrs**.
1.2.	**Microbiological techniques:** Sterilization, disinfection and antimicrobial agents, culturing techniques.	**2 hrs**.
Unit II		
2.1.1.	**Bacterial infections:** Infections associated with following Gram-positive bacteria – *Bacillus anthracis, Clostridium* spp., *Pneumococcus* spp., *Corynebacterium*spp., and, Streptococcal and Staphylococcal infections.	**2 hrs**.
2.1.2.	Infections associated with following Gram-negative bacteria –Species of *Salmonella, Shigella, Klebsiella, Proteus, Yersinia and Escherichia, Vibrio, Pseudomonas, Burkholderia, Neisseria, Haemophilus, Campylobacter, Bordetella, Brucella.*	**3 hrs**.
2.1.3.	Infections associated with Mycoplasmas, *Mycobacterium tuberculosis* and *M. leprae,* Spirochetes (Species of*Treponema, Borrelia*and*Leptospira),* Actinomycetes, Rickettsiae and Chlamydiae.	**3 hrs**.
Unit III	**Diagnostic Bacteriology:**	
3.1.	Collection of clinical samples (Sputum, Skin, Blood, Urine and Stool) with proper precautions. Method of transport of clinical samples to the laboratory and storage, handling, transport. General methods of isolation and identification of pathogenic bacteria from clinical specimens: Microscopic examination of a sample by staining - Gram staining, Ziehl-Neelsen staining, Giemsa staining; Culture methods - Preparation and use of culture media (Blood agar, Chocolate agar, Lowenstein-Jensen medium).	**6 hrs.**
3.2.	Growth and Biochemical Characteristics. Rapid Methods of Identification: Serological Methods - Agglutination, ELISA, Immunofluorescence; Nucleic acid-based methods -PCR, Nucleic acid probes. Kits for Rapid Detection of Bacterial Pathogens causing Typhoid, Syphilis and Cholera; Bacteriophage Typing; Molecular Methods and Analysis of Metabolic Products.	**6 hrs**.
3.3.	Testing for Antibiotic Sensitivity in Bacteria using disc diffusion method, Determination of resistance/sensitivity of bacteria against antibiotics (Penicillin/Streptomycin). Determination of minimal inhibitory concentration (MIC) of an antibiotic (Penicillin/ Streptomycin).	**6 hrs**.
Unit IV		
4.	Computers in Clinical Microbiology, Concepts of Quality Assurance in Diagnostics.	**2 hrs.**

Reference Books:

1. Ananthanarayan and JayaramPaniker. 2017. **Textbook of Microbiology**, 10th Edn. Universities Press.
2. Felix Klemperer and Ernst Levy.2018.**Elements of Clinical Bacteriology: For Physicians and Students.** Kessinger Publishing**.**
3. James Versalovic, Karen, C. C.2011. **Manual of Clinical Microbiology,** Vol. 1 & 2,10th Edn.ASM Press.
4. Keith Steuthers, J.2017.Clinical **Microbiology**, 2nd Edn.CRC Press.
5. Mandel, G.L. Bennet, J.E. and Dolin, R. 1995. **Principles and practice of infectious disease.** 4th Edn. Churchill Living stone. New York.
6. Richard, A. Harvey. 2001. Lippincott's Illustrated Reviews. Microbiology, 3rdEdn.
7. Samuel Baron. **Medical Microbiology**, 2nd Edn., Addison – Wesley Publication & Co., New York. 1986.

B.Sc. Microbiology CBCS Scheme, 2018 onwards
Paper code: 18MBSC-B2
Title of the paper: CLINICAL PARASITOLOGY
Format: 2 hrs per week lecture &/ laboratory period
Teaching hours- 32(2 hrs. /week)
No. of credits- 2

Course Objectives:
1. To understand the existence of pathogenic parasites in different environments.
2. To study some selected parasites human diseases, their diagnosis, treatment and control

Course Description:
The paper focuses on the essentials of clinical lab skills for microbiology. The course covers the area of Clinical Parasitology and Microbiology diagnostics including serodiagnostics and molecular diagnostics. Coursework may include lectures and training in a lab or hospital, as part of the curriculum, where the students have the opportunity to use state-of-the-art instrumentation. They will also meet for a few hours for reading, discussing and analyzing scientific articles from Clinical Parasitology journals as a partial requirement for internal assessment.

Course Learning Outcome(CO):

CO	After the completion of this course, the student will be able to	Cognitive Level
CO-01	Describe and perform various techniques used in the proper collection, transportation, and processing, microbiological analysis of the clinical specimens and accurate interpretation of results.	**Apply**
CO-02	Have a working knowledge of the principles of techniques performed	**Apply**
CO-03	Get prepared to pursue courses which lead them to careers such as clinical laboratory technologists, Clinical microbiologist, Research	**Apply**
CO-04	Appreciate the role of microbiology in relation to the professions in the Clinical laboratory set up and everyday life	**Understand**

SYLLABUS:

Unit I		
	Classification and Brief history of Protozoa and helminthic infections – mechanism of disease production by parasites.	**4 hrs.**
Unit II		
	Aetiology, life cycle, epidemiology, symptomatology, pathogenesis, lab diagnosis, treatment, prevention and control of: **Protozoan parasites** - *Entamoebahistolytica, Giardia lamblia, Plasmodium* spp., *Balantidium coli, Trypanosoma*spp., *Leishmania*spp., *Toxoplasma gondii, Trichomonasvaginalis.*	**8 hrs.**
Unit III		
	Brief introduction of Helminths - *Ascarislumbricoides, Enterobiusvermicularis, Trichuristrichuria, Ancylostoma duodenum and Necatoramericanus, Wucherriabancrofti, Taeniasolium&Taeniasaginata, Echinococcusgranulosus, Hymenolepsis nana, Schistosomahaematobium, Fasciola hepatica.*	**10 hrs.**

Unit IV		
4.1.	Mediators/vectors of parasitic infection. Recent advances in the diagnosis and control of parasitic infections. Newly emerging parasitic infections.	**5 hrs.**
4.2.	Collection and processing of clinical samples. Immuno- and molecular techniques. Detection of *Plasmodium* in blood. Detection of protozoa andhelminths in clinical samples and water. Staining methods.	**5 hrs.**

Reference Books:

1. Anne, Z. and Gary, C.2006. **Veterinary Clinical Parasitology**. Blackwell publishing.
2. Pearson, R.D., Gillespie, S.H. 2009. **Principles and Practice of Clinical Parasitology**, 1st Edn .Wiley, John & Sons.
3. Sun, T. 2012. **Progress in Clinical Parasitology**. Springer-Verlag New York, LLC.
4. Zeibig, E. 2012. **Clinical Parasitology: A Practical Approach**, 2nd Edn. Elsevier Health Sciences.

B.Sc. Microbiology CBCS Scheme, 2018 onwards

Paper code: 18MBSC-B3

Title of the Paper: PLANT CLINIC

Format: 2 hrs. per week lecture/ laboratory period/fieldwork

Teaching hours- 32(2 hrs. /wk.)

No. of credits- 2

Course Objectives:

1. To develop necessary skills for diagnosis of plant diseases and deficiencies and to differentiate them.
2. To acquaint the students with ecologically safe and environmentally sustainable methods for plant disease management.

Course Description:

The course includes lecture, laboratory experiments and fieldwork to cover topics on plant disease diagnosis, epidemiology and integrated disease and pest management methods. Techniques of isolation, culturing and microscopic examination of plant pathogens, testing of antimicrobial activity, formulation of biopesticides and biofertilizers will be introduced.

Course Learning outcome(CO):

CO	After the completion of this course, the student will be able to	Cognitive Level
CO-01	oriented and trained to work as local plant health extension specialists, known as plant doctors to help farmers struggling with plant pests and diseases	Apply
CO-02	provide diagnoses and management advice for mitigating crop loss and enhancing plant health and economic benefit.	Analyse

Unit I		
1.1.	Introduction to plant clinic and Plant Doctor Concept, objectives and need.	**1 hr.**
1.2.	Brief introduction to plant biology- anatomy, physiology and ecology- factors affecting plant growth (biotic and abiotic (edaphic) factors), the role of microorganisms in plant growth. Effect of pathogens on seed germination and vigour index (paper towel method).	**3 hrs.**
1.3.	Plant disease field survey, collection and assessment techniques: General principles, planning the survey, sampling procedures, sample size, sampling patterns and data collection, collection of diseased specimens.	**2 hrs.**
Unit II		
2.1.	Plant disease diagnosis**:** The basics of plant disease diagnosis, diagnostic process.	**1 hr.**
2.1.1.	Symptomatology: Field observation of plant diseases:	**2 hrs.**

	Symptoms of plant diseases due to fungi, bacteria, viruses, phytoplasmas, nematodes and phanerogamic parasites on agricultural and horticultural crops as well as forest nursery plants. Insect infestation. Post-harvest diseases: Study of pathogens in fruit from local market.	
2.1.2.	Symptoms of nutrient deficiencies in plants.	**1 hr.**
2.2.	Etiology: Isolation and identification of plant pathogens (farms, orchards,	
2.2.1.	Microscopy & Culture methods: Aseptic isolation of the pathogen from plant material (Roots, leaves and stem fragments) - Surface sterilization of diseased specimen, incubation in a moist chamber. Isolation of bacterial and fungal pathogens by Serial dilution and plating in culture media. Selective media for isolating plant pathogens, use of antibiotics and selective fungicides. Induction of sporulation- Modification of culture media, Near UV light exposure. Seed health testing by SBM- Isolation of fungi from locally available seed samples.	**3 hrs.**
2.2.2.	Molecular approaches to pathogen identification in plant material: Use of ELISA, Real-time PCR. Nucleic acid-based probes for detection of plant pathogens including non-culturable organisms. Identification of Viral Diseases.	**2 hrs.**
Unit III		
3	Management of Plant Diseases: Integrated Management. (Use of cultural, physical, chemical, biological methods of disease management). Disease management versus Disease control.	**2 hrs.**
3.1.	Cultural and Physical methods: Field sanitation, use of clean planting material, crop rotation, trap crops, time of sowing, planting distance and tillage. Exclusion, eradication, immunization & resistance development, direct protection, plant quarantine. Soil sterilization: Concept, objectives and methods.	**2 hrs.**
3.2.	Chemical methods: Fungicides- Definition, classification and ideal characteristics of fungicides, active ingredients, formulations, methods of application, mode of action, and uses. Plant protection equipment –pneumatic air pump, knapsack sprayer, and mist blower cum duster.	**2 hrs.**
3.2.1.	Sulphur fungicides: Inorganic – Wettablesulphur, Organic – Thiram. Copper fungicides- Bordeaux mixture. Mercuric chloride – Agrosan. Heterocyclic nitrogenous compounds – Captan. Benzene compounds – Dexon. Systemic fungicides – Bavistin and Vitavax. Natural Pesticides: Nicotine, Neem and Pyrethrum. Rodenticides – Zinc phosphide. Nematicides- Nemagon, Propoxur. Weedicides- 2, 4-D. Antibiotics – Streptomycin and Aureofungin	**3 hrs.**

3.2.2.	Seed treatment: Concept, objective, traditional and modern methods of seed treatment. Effect of fungicide on seed-borne pathogens.	**2 hrs.**
3.2.3.	Control measures and environment: Pollution due to chemicals, residual effects, toxicity, antidote, safety measures. Symptoms of poisoning and use of antidotes, and precautions in using pesticides.	**2 hrs.**
3.3.1.	Biological control- Definition, need examples and role. Preparation of *Trichoderma*culture and formulation of the biocontrol agent. Use of Pheromones.	**1 hr**.
3.3.2.	Biotechnological Methods of Plant disease Management: Genetic engineering and crop protection. Cross Protection. Gene Silencing and disease control- Mechanism of gene silencing and control of viral diseases. Engineered resistance to viral, bacterial, fungal and insect diseases of crop plants. Genetically modified organisms (GMOs) - Bt Cotton.	**3 hrs**.
Unit IV		
4.1.	A collection of plant disease specimens (5no.s; 1 bacterial, 3 fungal and 1 viral) to be submitted for internal assessment.	
4.2.	Fieldtrips- reports to be submitted for internal assessment.	

Reference Books:

1. Agrios, G N. 2005. **Plant Pathology,** 5th Edn. Elsevier.
2. Johnston, A., Both, C.1983. **Plant Pathologist's Pocketbook**, 2nd Edn. Commonwealth Mycological Institute, Oxford and IBH Pub. Co. Calcutta.
3. Mehrotra, R. S. 1983. **Plant Pathology.** Tata McGraw Hill Pub. Co. Ltd., New Delhi.
4. Nene, Y.L. and Thapliyal, P.N.1979. **Fungicides in Plant Disease Control**, 2nd Edn. Oxford & IBH Publishing Co., New Delhi.
5. Rangaswamy, G. and Mahadevan, A.2002.**Diseases of Crop Plants in India**. Prentice-Hall of India Pvt. Ltd., New Delhi.
6. Singh, R S .1973. **Plant Disease**. Oxford and IBH Pub.Co. New Delhi.
7. Vidhyasekaran, P. 2004. **Encyclopaedia of Plant Pathology**. Viva Books Pvt. Ltd., New Delhi.
8. Waller, J.M., Lenné, J.M. and Waller, S.J. (Editors).2002. **Plant Pathologist's Pocketbook**, 3rd Edn. CABI Publishing.

B.Sc. Microbiology CBCS Scheme, 2018 onwards

Paper code: 18MBSC-B4
Title: PROJECT WORK WITH SEMINAR
No. of credits- 2

Course Objectives:

1. To orient students towards research.
2. To inculcate qualities such as independent thinking, creativity and self-motivation.

Course description:

It involves a short term project work / Dissertation of a minimum of one-month duration to be carried out by the students under the supervision of the teacher(s) in the college or a guide in any research institute. The topic of the project would be selected by each student in consultation with the guide. The project must be outlined and proposed to the guide for approval.

The Dissertation of the student must include Objectives, Review of Literature, Methodology, Results and Discussion. It should be approved and certified by the guide, and the final dissertation submitted before the last week of the course.

Grading would be based on a continuous evaluation that would include punctuality, hard work, record keeping, intellectual inputs, data presentation, interpretation etc.

The student has to present a seminar, at the time of final submission of the dissertation, which may be evaluated by external and internal examiners.

Course Learning outcome(CO):

CO	After the completion of this course, the student will be able to	Cognitive Level
CO-01	To appreciate the importance of research in the development of Science	Apply
CO-02	To get exposed to the vast opportunities available to pursue a career in biological research	Analyse

B.Sc. Microbiology CBCS Scheme, 2018 onwards
Paper code: 18MBSC-B5
Title: ENVIRONMENTAL MICROBIOLOGY AND PUBLIC HEALTH
Format: 2 hrs per week lecture &/ laboratory period
Teaching hours- 32(2 hrs. /week)
No. of credits- 2

Course Objectives:

1. To create awareness about the diversity of microbial agents, including bacteria, archaea, viruses, fungi, microsporidia, in the environment that can impact public health
2. To understand the effect of microorganisms on the global environment e.g. their role in the cycling of elements and factors that limit microbial growth and survival in natural habitats
3. To learn the methods for studying microbial populations and their function in the natural environment
4. To create awareness about the methods for fast and efficient identification of pathogenic/allergenic microorganisms and biological material in environmental media, which can be used in appropriate tools and models for the protection of public health.
5. To learn the application of microbial ecology principles for industrial, environmental or public health benefits

6. To elaborate on treatment technologies that are available to mitigate the risks to public health.

Course description:
An introduction to the diversity of microbial agents that can impact the public health and environmental systems, the course focuses on the microbial hazards found in waters, soils and air. Molecular biology techniques and the current regulatory methods for investigating pathogens and the surrogate indicators will be discussed. Treatment and engineering strategies are discussed. The course also provides an introduction to Quantitative Microbial Risk Assessment (QMRA). The concepts related to exposure assessment and risk characterization will be included.

Course Learning outcome(CO):

CO	After the completion of this course, the student will be able to	Cognitive Level
CO-01	List and explain the biological principles required to understand the distribution of infectious and non-infectious diseases of public health importance, the current research related to public health problems, and the methods for their prevention and control	Apply
CO-02	Understand the application of biological principles in the management of diseases of public health significance.	understand
CO-03	Highlight areas of public health where recent biological research is likely to be of particular importance.	analyse

.
SYLLABUS:

Unit I		
1.1.	**Basic concepts of Ecology and Environment**: Concept and components of Ecosystem.	**2 hrs.**
1.2.	Environmental factors influencing the growth and survival of microorganisms. Physical factors - temperature, light, osmotic pressure and hydrostatic pressure. Chemical factors - pH, O_2 and CO_2. Biological factors - Interactions of microbial population and community dynamics.	**2 hrs.**
Unit II		
2.1.	**Importance of Microorganisms in the environment:** Faecal pollution of water and water-borne diseases, bioaerosols and air-borne diseases.	**3 hrs.**
2.2.	Biotechnological approaches for the management of global environmental problems (ozone depletion, greenhouse effect and acid rain). Xenobiotic metabolism, biomagnification.	**2 hrs.**
2.3.	Microorganisms and climate change.	**2 hrs.**
Unit III		
3.1.	**Interrelationship of Environment and Public health:** Concept and definition of health, Determinants of health. Responsibility for health-	**3 hrs.**

	Individual, Community, State and International responsibility. Indicators of health. Causation of infectious disease.	
3.2.	Definition of terms:-infection, contamination, infestation, host, infectious disease, communicable disease, epidemic, pandemic, endemic, sporadic, exotic, zoonotic, epizootic, enzootic and epornithic diseases.	**2 hrs.**
3.3.	Environment (Indoor and outdoor) and its effect on Public health, epidemiology. Environmental health hazards-Sources and prevention. Sources of infection and allergies. Modes of transmission of infection. Reservoirs, carriers and vectors of communicable diseases. Role of WHO in pandemic alerts. Quantitative Microbial Risk Assessment (QMRA)-WHO guidelines.	**5 hrs.**
Unit IV		
4.1.	**Management of health care waste:** Definition of waste, Types of health care waste- Infectious and non-infectious. Sources of infectious agents in health care waste. Categories of health care waste as per WHO. Impact (Health hazards and risks associated) of infectious waste on the environment and public health.	**3 hrs**.
4.2.	Principles of managing different categories of waste- collection, handling, segregation, colour coding, storage, transportation, treatment and disposal.	**2 hrs.**
Unit V		
5.1.	**Biosafety:** Introduction to biosafety, definition, Objectives of safety guidelines, Assessment of risk during laboratory research- biosafety levels, Containment – Physical containment, Biological containment.	**3 hrs.**
5.2.	Risk assessment for biotechnology products: Release of GMO's to the environment and their impact, ethical issues. Biosafety during industrial production using GMO's. Biosafety guidelines in India**.**	**3 hrs.**

Reference Books:

1. Atlas, R.M. and Bartha, R. 2000. **Microbial Ecology**: **Fundamentals & Applications**, 4thEdn. Benjamin/Cummings Science Publishing, USA
2. da Silva, N., Taniwaki, M.H., Junqueira, V.C., Silveira, N., Nascimento, M.S., Gomes, R.A.R. 2012. **Microbiological Examination Methods of Food and Water. A Laboratory Manual.** CRC Press
3. Maier, R.M., Pepper, I.L. and Gerba, C.P. 2009. **Environmental Microbiology,** 2nd Edn., Academic Press
4. Hurst, C.J., Crawford, R.L., Garland, J.L., Lipson, D.A. 2007. **Manual of Environmental Microbiology**, 3rd Edn., ASM Press

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B.Sc. Microbiology CBCS Scheme, 2018 onwards
Paper code: 18MBSC-B6
Title: MARINE MICROBIOLOGY:
Format: 2 hrs per week lecture &/ laboratory period
Teaching hours- 32(2 hrs. /week)
No. of credits- 2

Course objectives:

1. To study the distribution of various types of microorganisms in the marine environment.
2. To understand the importance of marine microorganisms.
3. To study the effect of pollution on marine fauna and flora.
4. To create awareness about the applications of marine microbiology.
5. To provide the students with a basic theoretical and practical understanding of the interactions between microorganisms and ocean processes and discuss their future role in a rapidly changing planet.

Course Description:

Marine microbiology is the study of the microorganisms (bacteria, archaea, viruses and microbial eukaryotes) in the marine environment, including their biodiversity, ecology and biogeochemistry. In the past decade, marine microbiology has emerged as an important and dynamic discipline. Significant advances made in this exciting field are leading to a new understanding of biodiversity, ecology and biogeochemistry.

Course Learning Outcomes (CO):

CO	After the completion of this course, the student will be able to	Cognitive Level
CO-01	Gain insight into the diversity of microorganisms in the marine environment	Understand
CO-02	Devise experimental strategies for analyzing microbial populations, their activity and applications	create
CO-03	Account for the role and significance of marine microorganisms	analyse

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SYLLABUS:

Unit I		
1.1.	**Introduction to Marine Microbiology**: History of Oceanography, marine biological institutions.	**2 hrs.**
1.2.	Zonation and microbial stratification in the marine environment. Effect of movement of water in the marine environment.	**2 hrs.**
1.3.	Laws of ecology with particular reference to the marine ecosystem: environmental factors (biotic and abiotic) and their influence on the distribution	**3 hrs.**

	of microorganisms. The microbial food web in the marine environment.	
Unit II		
2.1.	Marine microorganisms: Diversity of microorganisms - Archaea, bacteria, cyanobacteria, algae, fungi, viruses and actinomycetes. Plankton- classification based on size, mode of life and habitat.	**3 hrs.**
2.2.	Microbial life in the deep sea bed. Marine extremophiles- halophiles, barophiles, thermophiles, psychrophiles.	**3 hrs.**
2.3.	Enumeration of marine bacteria: sampling and samplers, processing and actual enumeration procedures (culture-dependent and culture-independent methods/ molecular methods: RFLP, FISH). Phytoplankton and Zooplankton - methods of collection and detection; phytoplankton detection by remote sensing.	**4 hrs.**
2.4.	Sea as a biological environment. Ecology of coastal, shallow and deep-sea microorganisms. UV radiation effects on microbes and microbial processes, uptake and regeneration of inorganic nutrients by marine heterotrophic bacteria, bacterivory: interactions between bacteria and their grazers, symbiosis and mixotrophy among pelagic microorganisms, marine viruses and their ecological impact.	**5 hrs.**
Unit III		
3.1.	Activities of marine microorganisms and their significance: Phytoplankton photosynthesis and primary production, biogeochemical cycling of C, N, S & P. Bioluminescence, antimicrobial activity, biofilm formation, polymer and surfactant production, toxin production.	**4 hrs.**
3.2.	Advantages and disadvantages of marine microorganisms including their importance in marine biotechnology- marine yeasts. Marine microbes - drug discovery, biofuel sources.	**2 hrs.**
3.3.	Marine pollution microbiology; Microbial biodegradation of natural and synthetic material in the marine environment- pesticide, cellulose degradation, hydrocarbon production. Bioremediation of xenobiotics (pesticides, plastics), oil, heavy metals. Mining and metal biotechnology. Some common diseases of marine fauna.	**4 hrs.**

Reference Books:

1. Atlas, R.M. 1988. Microbiology- Fundamentals and Applications, International Edn. Maxwell McMillan
2. Chauhan. 2009. Cyanobacteria- Antimicrobial Activity. NIPA.
3. Horst, D. S. and Matthias, Z. 2006. Marine Geochemistry. Springer
4. Kirchman, L. 2000. Microbial Ecology of the Oceans. John Wiley and Sons.
5. Munn, C. 2011. Marine Microbiology: Ecology & Applications. Taylor & Francis Press
6. Paul, F. and Andrew, H. K. 2007. Evolution of Primary Producers in the Sea. Academic Press.
7. Rheinheimer, G. 1980. Aquatic Microbiology-an Ecological Approach. Blackwell Scientific Publications
8. Scheper, T. 2011.Marine Biotechnology. Springer-Verlag, New York, LLC

B.Sc. Microbiology DSE & SEC

Time: 2 Hours **Max. Marks: 30**

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PART A		
Q.1. **a)** **b)**	**Answer the following:** Long answer type question OR Long answer type question	**1X10=10**
PART B		
Q.2.	**Write short notes on any THREE of the following:**	**3X5= 15**
a)	Short answer question	
b)	Short answer question	
c)	Short answer question	
d)	Short answer question	
PART C		
Q.3.	**Define/Expand the following: (Objective type questions)**	**5X1=05**
a)		
b)		
c)		
d)		
e)		

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